DBMS – Sales and Customer Analytics System

Project Title: Big Basket E-commerce Database Management System

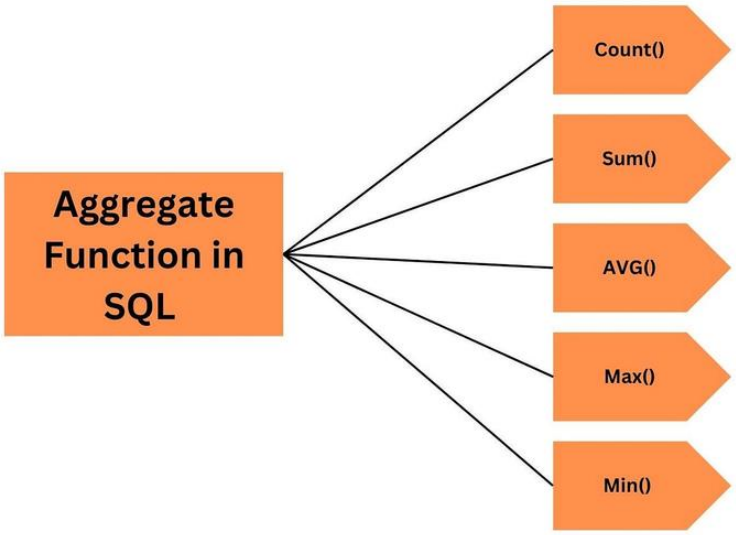
OBJECTIVE

1. Apply COUNT(), SUM(), AVG(), MIN(), MAX().
2. Generate total sales reports.
3. Find top customers based on purchase amount.
4. Identify best-selling products.
5. Perform category-wise sales analysis.

1. Aggregate Functions

Definition:

Aggregate functions are SQL functions used to perform calculations on a group of rows and return a single result.



1.1 COUNT() :

Find the total number of orders

```
Select Count(*)as Total_Orders
from Orders;    Syntax Error at

TOTAL_ORDERS
------------
          10
```

1.2 SUM():

Find the total sales amount

```
Select Sum(Total_Amount) as Total_Sales
from Orders;

TOTAL_SALES
-----------
       39250
```

1.3 AVG():

Find the average order amount

```
Select Avg(Total_Amount) as Average_Sales
from Orders;

AVERAGE_SALES
-------------
         3925
```

1.4 MIN():

Find the lowest order amount

```
Select min(Total_Amount) as Minimum_Sales
from Orders;

MINIMUM_SALES
-------------
          500
```

1.5 MAX():

Find the maximum order amount

```
Select max(Total_Amount) as Maximum_Sales
from Orders;

MAXIMUM_SALES
-------------
        28000
```

2. Generate Total Sales Reports

A sales report can be generated using the SUM() function on the Total Amount column.

```
Select Count(Order_ID) as Total_Orders, Sum(Total_Amount) as Total_Sales, Avg(Total_Amount) as Average_Sales, Min(Total_Amount) as Minimum_Sales,
Max(Total_Amount) as Maximum_Sales
from Orders;

TOTAL_ORDERS TOTAL_SALES AVERAGE_SALES MINIMUM_SALES MAXIMUM_SALES
------------ ----------- ------------- ------------- -------------
          10       39250          3925           500         28000
```

2.1 Customer-wise Sales Report

To find how much each customer has purchased

```
SELECT
    c.Customer_ID,
    c.Customer_Name,
    SUM(o.Total_Amount) AS Total_Purchase
FROM Customer c
JOIN Orders o
    ON c.Customer_ID = o.Customer_ID
WHERE o.Order_Status <> 'Cancelled'
GROUP BY
    c.Customer_ID,
    c.Customer_Name
ORDER BY Total_Purchase DESC;

CUSTOMER_ID CUSTOMER_NAME                                          TOTAL_PURCHASE
----------- ---------------------------------------------------- --------------
        109 Rohan Gupta                                                   28000
        101 Arun Kumar                                                     3150
        107 Karan Mehta                                                    2500
        103 Rahul Singh                                                    1800
        102 Priya Sharma                                                   1250
        105 Vikram Patel                                                    850

6 rows selected.
```

3. Find Top Customers Based on Purchase Amount

The SUM() function can be combined with GROUP BY and ORDER BY to identify customers with higher purchase amounts

```
Select
    c.Customer_ID,
    c.Customer_Name,
    COUNT(o.Order_ID) AS Total_Orders,
    SUM(o.Total_Amount) AS Total_Purchase
from Customer c
join Orders o
    on c.Customer_ID = o.Customer_ID
Group by
    c.Customer_ID,
    c.Customer_Name
Order by Total_Purchase DESC;
```

```
CUSTOMER_ID CUSTOMER_NAME                                          TOTAL_ORDERS
----------- -------------------------------------------------- ------------
TOTAL_PURCHASE
--------------
        109 Rohan Gupta                                                   1
      28000

        101 Arun Kumar                                                    3
       3150

        107 Karan Mehta                                                   1
       2500


CUSTOMER_ID CUSTOMER_NAME                                          TOTAL_ORDERS
----------- -------------------------------------------------- ------------
TOTAL_PURCHASE
--------------
        103 Rahul Singh                                                   1
       1800

        102 Priya Sharma                                                  1
       1250

        106 Anjali Das                                                    1
       1200


CUSTOMER_ID CUSTOMER_NAME                                          TOTAL_ORDERS
----------- -------------------------------------------------- ------------
TOTAL_PURCHASE
--------------
        105 Vikram Patel                                                  1
        850

        104 Sneha Reddy                                                   1
        500


8 rows selected.
```

4. Identify Best-Selling Products

The Order Details table contains the quantity of each product ordered. Therefore, we can use SUM(Quantity) to find the total quantity sold for each product.



```
Select
p.Product_ID,
p.Product_Name,
Sum(od.Quantity) as Total_Quantity_Sold
from Product p
join Order_Details od
on p.Product_ID = od.Product_ID
Group by
p.product_ID,
p.Product_Name
Order by Total_Quantity_Sold DESC;

PRODUCT_ID PRODUCT_NAME
---------- ----------------------------------------
TOTAL_QUANTITY_SOLD
-------------------
       104 Keyboard
                 3

       103 Tablet
                 2

       101 Laptop
                 2


PRODUCT_ID PRODUCT_NAME
---------- ----------------------------------------
TOTAL_QUANTITY_SOLD
-------------------
       105 Mouse
                 2

       107 Printer
                 2

       110 Headphones
                 2


PRODUCT_ID PRODUCT_NAME
---------- ----------------------------------------
TOTAL_QUANTITY_SOLD
-------------------
       109 Webcam
                 1

       106 Monitor
                 1

       108 Speaker
                 1


PRODUCT_ID PRODUCT_NAME
---------- ----------------------------------------
TOTAL_QUANTITY_SOLD
-------------------
       102 Smartphone
                 1


10 rows selected.
```

4.1 Product Sales Amount

Instead of only calculating quantity, we can calculate the total sales generated by each product.

```sql
Select
p.Product_ID,
p.Product_Name,
Sum(od.Quantity) as Total_Quantity_Sold,
Sum(od.Sub_total) as Total_Sales
from Product p
join Order_Details od
on p.Product_ID = od.Product_ID
Group by
p.Product_ID,
p.Product_Name
Order by Total_Sales DESC;
```

```
PRODUCT_ID PRODUCT_NAME
---------- ------------------------------------
TOTAL_QUANTITY_SOLD TOTAL_SALES
------------------- -----------
       101 Laptop
                 2       70000

       102 Smartphone
                 1       45000

       107 Printer
                 2       32000


PRODUCT_ID PRODUCT_NAME
---------- ------------------------------------
TOTAL_QUANTITY_SOLD TOTAL_SALES
------------------- -----------
       110 Headphones
                 2       28000

       105 Mouse
                 2       22000
```

```
       103 Tablet
                  2       18000


PRODUCT_ID PRODUCT_NAME
---------- ----------------------------
TOTAL_QUANTITY_SOLD TOTAL_SALES
------------------- -----------
       104 Keyboard
                  3       15500

       109 Webcam
                  1       15000

       106 Monitor
                  1       12500


PRODUCT_ID PRODUCT_NAME
---------- ----------------------------
TOTAL_QUANTITY_SOLD TOTAL_SALES
------------------- -----------
       108 Speaker
                  1        8500


10 rows selected.
```

5. Category-wise Sales Analysis

Category-wise analysis helps determine the sales generated by each product category.

```
Select
c.Category_ID,
c.Category_Name,
Sum(od.Quantity) as Total_Quantity_Sold,
Sum(od.Sub_total) as Total_Sales
from Category c
join Product p
on c.Category_ID = p.Category_ID
join Order_Details od
on p.Product_ID = od.Product_ID
Group by
c.Category_ID,
c.Category_Name
Order by Total_Sales DESC;

CATEGORY_ID CATEGORY_NAME                    TOTAL_QUANTITY_SOLD TOTAL_SALES
----------- -------------------------------- ------------------- -----------
          1 Electronics                                      17      266500
```

6. Combined Business Sales Report

A complete sales report can combine customer, order, product, and category information.

```
Select
c.Customer_Name,
o.order_ID,
p.Product_Name,
cat.Category_Name,
od.Quantity,
od.Sub_total
from Customer c
join Orders o
on c.Customer_ID = o.Customer_ID
join Order_Details od
on o.Order_ID = od.Order_ID
join Product p
on od.Product_ID = p.Product_ID
join Category cat
on p.Category_ID = cat.Category_ID
Order by o.Order_ID;
```

```
CUSTOMER_NAME                                          ORDER_ID
------------------------------------------------------ ----------
PRODUCT_NAME
--------------------------------------------------
CATEGORY_NAME                  QUANTITY  SUB_TOTAL
------------------------------ ---------- ----------
Arun Kumar                                          1001
Laptop
Electronics                             2      70000

Arun Kumar                                          1001
Keyboard
Electronics                             2      10500
```

```
CUSTOMER_NAME                                          ORDER_ID
-------------------------------------------------- ----------
PRODUCT_NAME
--------------------------------------------------
CATEGORY_NAME                  QUANTITY  SUB_TOTAL
------------------------------ ---------- ----------

Priya Sharma                                             1002
Keyboard
Electronics                             1       5000

Rahul Singh                                              1003
Smartphone

CUSTOMER_NAME                                          ORDER_ID
-------------------------------------------------- ----------
PRODUCT_NAME
--------------------------------------------------
CATEGORY_NAME                  QUANTITY  SUB_TOTAL
------------------------------ ---------- ----------
Electronics                             1      45000

Arun Kumar                                               1004
Tablet
Electronics                             2      18000

Sneha Reddy                                              1005

CUSTOMER_NAME                                          ORDER_ID
-------------------------------------------------- ----------
PRODUCT_NAME
--------------------------------------------------
CATEGORY_NAME                  QUANTITY  SUB_TOTAL
------------------------------ ---------- ----------
Monitor
Electronics                             1      12500
```

```
Vikram Patel                                                1006
Printer
Electronics                              2      32000


CUSTOMER_NAME                                          ORDER_ID
-------------------------------------------------- ----------
PRODUCT_NAME
--------------------------------------------------
CATEGORY_NAME                    QUANTITY  SUB_TOTAL
------------------------------ ---------- ----------
Anjali Das                                              1007
Speaker
Electronics                              1       8500

Arun Kumar                                              1008
Mouse
Electronics                              2      22000

CUSTOMER_NAME                                          ORDER_ID
-------------------------------------------------- ----------
PRODUCT_NAME
--------------------------------------------------
CATEGORY_NAME                    QUANTITY  SUB_TOTAL
------------------------------ ---------- ----------

Karan Mehta                                             1009
Webcam
Electronics                              1      15000

Rohan Gupta                                             1010
Headphones
```

```
CUSTOMER_NAME                                          ORDER_ID
-------------------------------------------------- ----------
PRODUCT_NAME
--------------------------------------------------
CATEGORY_NAME                    QUANTITY  SUB_TOTAL
------------------------------ ---------- ----------
Electronics                              2      28000


11 rows selected.


commit;
Commit complete.
```